Self-Supervised Learning

[Partly based on Andrew Zisserman's slides]



Self-Supervised Learning

Notes

- Infants learn mostly without labelling!
- Certain parts of the brain are mostly fixed in the early stages of life
  - "children up to age 18 months have similar vision to adults"
- Otherwise, the brain are quite plastic (live-ware)
- It could be possible to learn basic processing of sensory input without direct supervision

This is what self-supervised learning is trying to do (and in fact quite successfully)



[David Eagleman, "LIVEWIRED"]

["Visual development in children aged 0 to 6 years"]

Self-Supervised Learning

“Definition” and General Approach

- Self-Supervised Learning: Learn some (large) parts of the model parameters on merely input data (without any additional human-annotated labels)

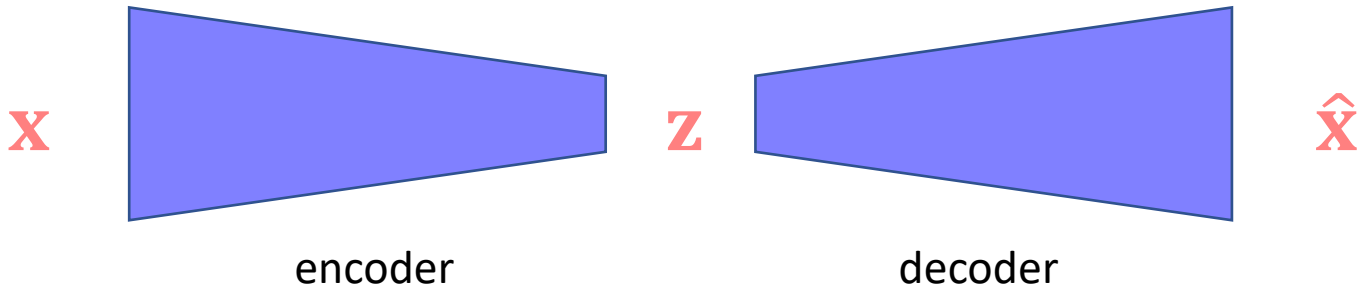
- General Idea:
  - Creatively produce supervised tasks from the mere input itself
  - Transfer the model either representation-level or parameter level



Simplest Case

Sample reconstruction

- Learn a “summarization” or “feature extraction” network
- Reconstruct the original input from the “summary” or the “feature”



There have been many creative examples of this in the recent years, which are quite successful